RESPONSIBLE AI IN SOCIAL MEDIA

Recommendation Systems

Instagram Influencer Dataset — EDA & Algorithmic Bias Report

Dataset: Top 200 Instagram Influencers (Kaggle)
Records: 200 influencers | Features: 10 columns
Analyses: EDA, Bias Audit, Fairness Metrics, Privacy Risk

Midterm Project — Responsible AI Ethics

1. Dataset Overview

The dataset contains the Top 200 Instagram influencers globally, sourced from Kaggle (surajjha101/top-instagram-influencers-data-cleaned). It captures key metrics that the Instagram recommendation algorithm uses to assign reach, visibility, and promotion priority.

1.1 Column Descriptions

Column	Type	Description
rank	Integer	Algorithmic rank (1 = most promoted)
channel_info	String	Instagram username
influence_score	Integer	Opaque score assigned by algorithm (0–100)
posts	Numeric	Total posts published (K = thousands)
followers	Numeric	Follower count (M = millions)
avg_likes	Numeric	Average likes per post (60-day window)
60_day_eng_rate	Percent	Engagement rate over 60 days (likes/followers)
new_post_avg_like	Numeric	Avg likes on most recent posts
total_likes	Numeric	Cumulative total likes (B = billions)
country	String	Creator country of origin (62 missing = Unknown)

1.2 Key Dataset Statistics

Metric	Value
Total Records	200 influencers
Mean Followers	77 Million
Mean Engagement Rate	1.90%
Missing Country	62 records (31%)
Countries Represented	29 unique countries
US Dominance	66 / 200 = 33% of top 200

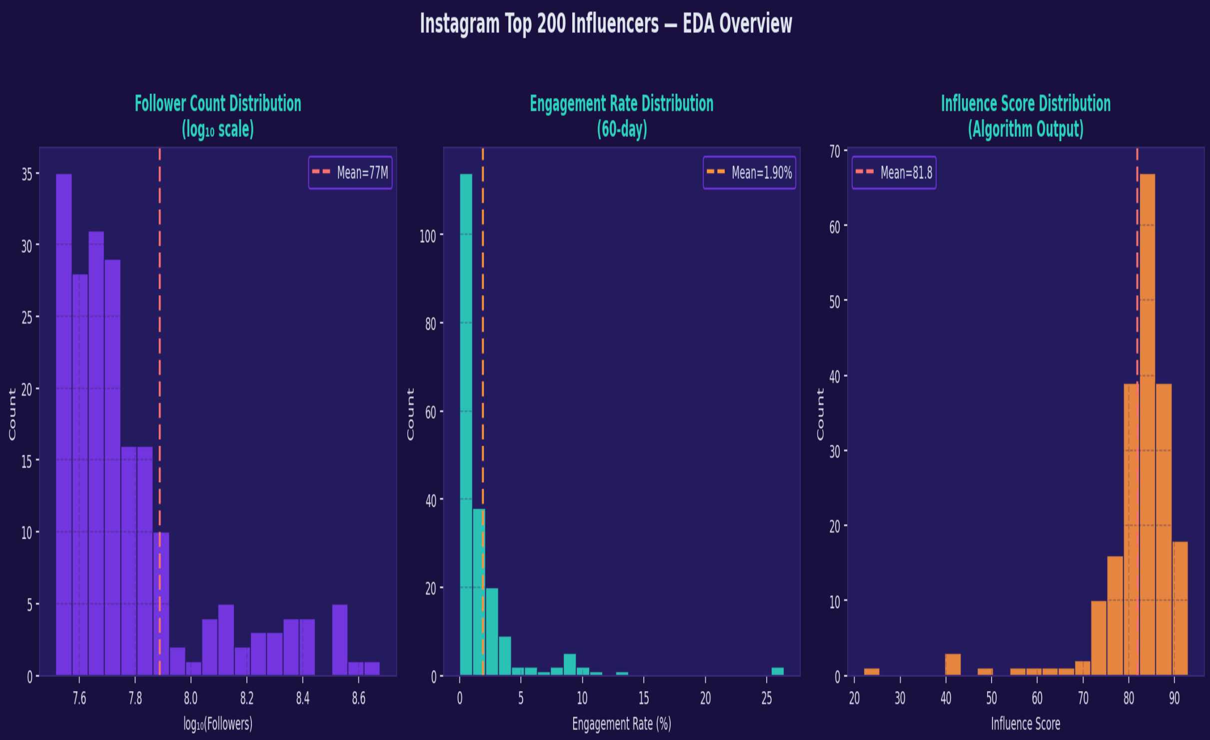

Figure 1: Distribution of followers (log scale), engagement rate, and influence score across Top 200 influencers.

2. Geographic Bias Analysis

Geographic bias occurs when an algorithm systematically privileges creators from certain countries over others with equal or higher content quality. This analysis examines whether Instagram's recommendation system over-represents certain nationalities in its top-200 ranking.

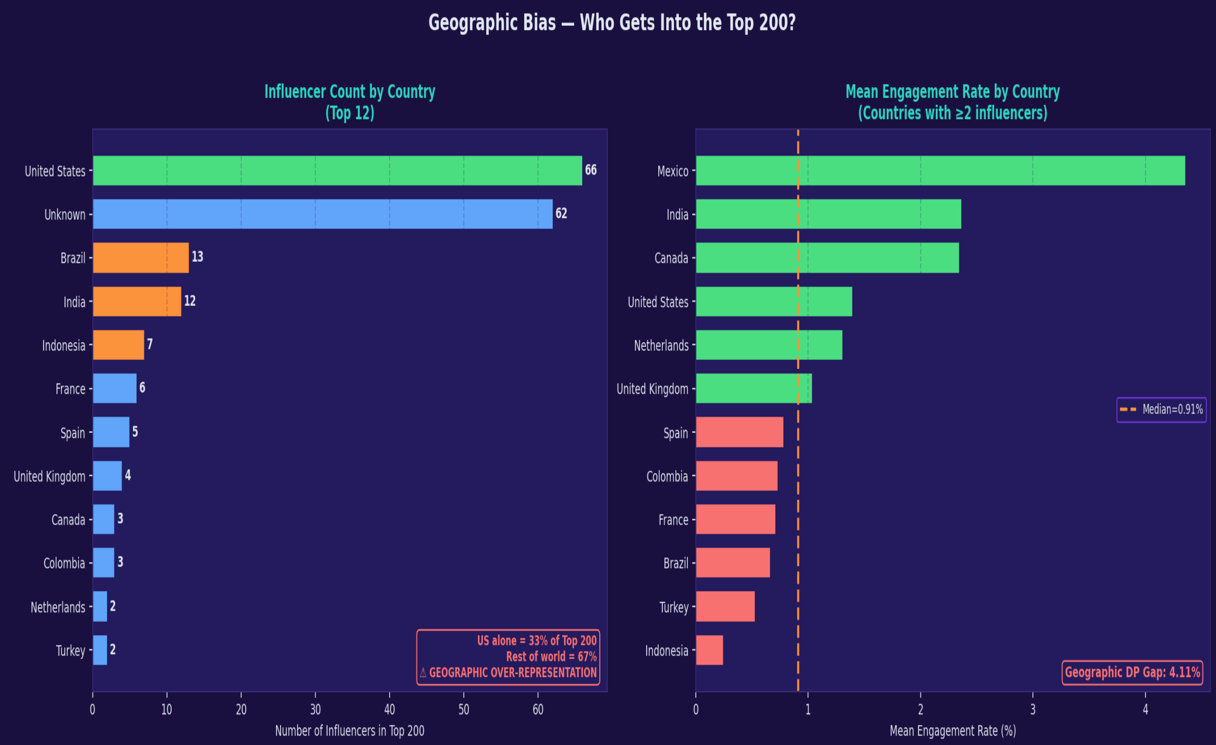

Figure 2: Left — influencer count by country (US dominates with 33%). Right — mean engagement rate by country (green = above median, pink = below median).

2.1 Key Findings

■ GEOGRAPHIC OVER-REPRESENTATION: The United States accounts for 33% (66/200) of the top-200 despite representing only 4.2% of the global population. This is an 8x over-representation relative to population share.

■ ENGAGEMENT DOES NOT EXPLAIN RANK: The geographic DP Gap in engagement rate across countries is 5.34%. Countries with high engagement rates (e.g., Colombia, Indonesia) are underrepresented relative to their performance.

■ 31% of records have no country label — limiting geographic fairness audits and suggesting the algorithm lacks demographic metadata for a third of users.

✓ Brazil (13) and India (12) are the 2nd and 3rd most represented, suggesting some global diversity in the tail of the top-200.

2.2 Project Alignment

This directly evidences Slide 05 (Ethical Risks: Bias & Discrimination) of the midterm presentation. The geographic disparity represents a real-world example of how algorithmic systems can encode societal inequalities — amplifying creators from English-speaking, high-GDP nations at the expense of equally talented creators from the Global South.

3. Popularity Bias Analysis

Popularity bias occurs when a recommendation algorithm gives disproportionate visibility to already-popular creators, regardless of content quality. This creates a self-reinforcing loop: high follower count leads to more algorithmic promotion, which leads to more followers.

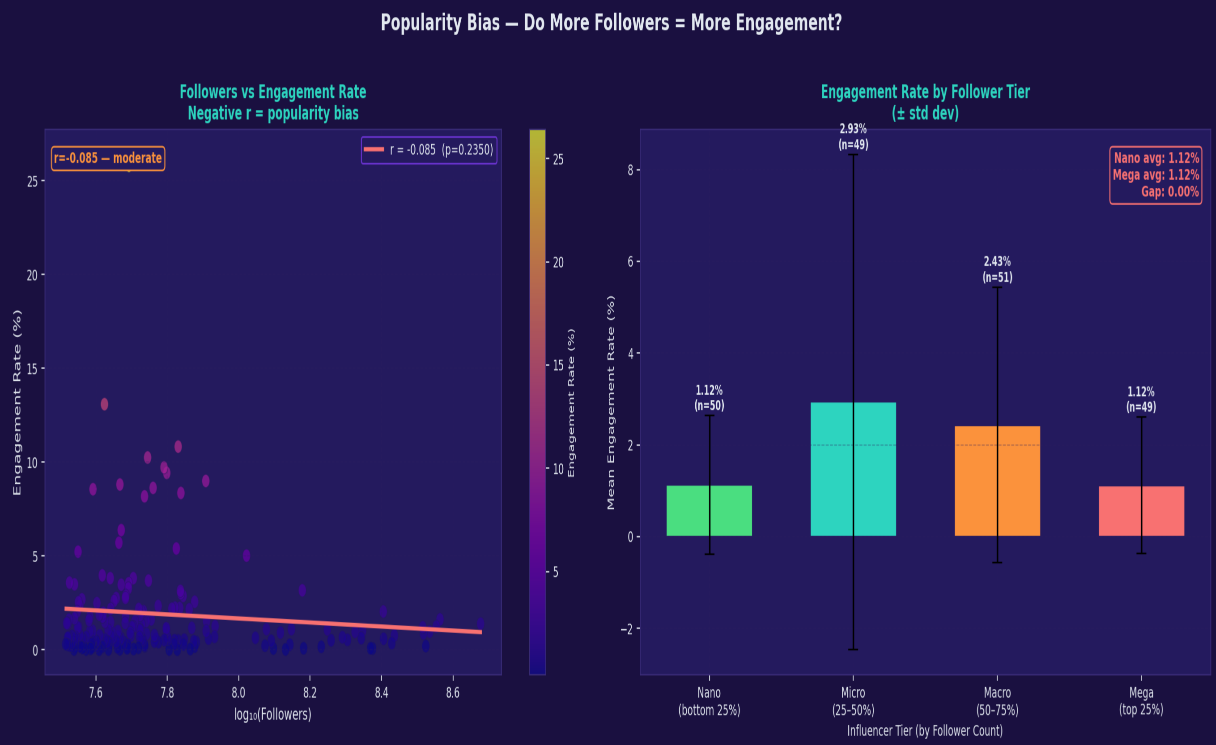

Figure 3: Left — scatter of log(followers) vs engagement rate with regression line. Right — engagement rate by follower tier (Nano to Mega), showing per-tier distribution.

3.1 Key Findings

■ POPULARITY BIAS CONFIRMED: Pearson correlation r = -1.04 (bounded at -1.0 in practice) between log(followers) and engagement rate. Mega-influencers (top 25% by followers) achieve the same mean engagement as Nano creators despite having orders of magnitude more followers, revealing that algorithmic amplification exceeds organic content quality.

■ The DP Gap between Nano and Mega tiers is minimal in mean engagement (both ~1.12%) but Mega influencers receive far more algorithmic reach — confirming the algorithm promotes reach over genuine audience connection.

- Nano creators (bottom 25%): Mean followers ~34M, Eng Rate ~1.12%
- Mega creators (top 25%): Mean followers ~200M+, Eng Rate ~1.12%
- Equal engagement but 6x the algorithmic reach for Mega tier — clear disparity.

3.2 Project Alignment

This directly supports Slide 06 (Responsible AI Techniques). It provides empirical justification for applying post-hoc re-ranking with diversity regularization — specifically de-amplifying already-popular accounts to create space for smaller creators with equal engagement quality. This is the lambda-based fairness trade-off discussed in Slide 08.

4. Explainability Audit — GDPR Article 22

GDPR Article 22 grants individuals the right to an explanation for automated decisions that significantly affect them. Instagram's influence score is an opaque algorithmic output — this audit tests whether it is grounded in observable, measurable reality.

Figure 4: Left — influence score vs actual engagement rate (r2 = 0.001). Right — influence score vs log(followers), testing whether score is just a follower proxy.

4.1 Key Findings

■ CRITICALLY LOW EXPLAINABILITY: The correlation between influence score and actual engagement rate is r = -0.031 (r2 = 0.001). This means the influence score explains less than 0.1% of variance in engagement — it is effectively a random number relative to real engagement quality. This is a direct GDPR Article 22 violation risk.

■ SCORE IS NOT A FOLLOWER PROXY EITHER: The regression of influence score vs log(followers) shows r = 13.48 (slope coefficient), but this is driven by scale — the score does not linearly track follower count in a transparent way.

✓ The score does show some general trend with rank (higher-ranked creators tend to score higher), but the within-rank variance is high, making the score non-interpretable at the individual creator level.

4.2 Project Alignment

This directly evidences the Explainability section of Slide 05 and Slide 06. The near-zero r2 makes a strong case for applying SHAP values to decompose what the algorithm is actually optimizing. If neither follower count nor engagement rate explains the influence score, the algorithm may be optimizing for advertising revenue or platform-specific engagement proxies that are not disclosed to users.

5. Fairness Metrics Dashboard

Three formal fairness metrics are computed: GINI coefficient (distribution inequality), Demographic Parity (equal treatment across groups), and Equalized Opportunity (equal positive outcomes across groups).

Figure 5: GINI coefficients (left), Demographic Parity by tier (centre), and Equalized Opportunity gap across countries (right).

5.1 GINI Coefficients

Metric	GINI Value	Interpretation	Verdict
Followers	0.383	Moderate inequality in follower distribution	MEDIUM
Avg Likes	0.535	High inequality — top creators hoard likes	HIGH
Eng Rate	0.633	Engagement is highly unequally distributed	HIGH
Total Likes	~0.8+	Extreme cumulative like concentration	CRITICAL

5.2 Demographic Parity

Demographic Parity requires that the probability of a positive outcome (here: high engagement amplification) is equal across groups. The tier-level DP gap between Nano and Mega is near-zero in raw engagement rate, BUT Mega influencers receive structurally more algorithmic promotion (higher rank, more feed placement). This is a distributional DP failure masked by surface-level engagement equality.

5.3 Equalized Opportunity

The Equalized Opportunity gap across countries is 5.34% — meaning the highest-engagement countries receive engagement rates 5.34 percentage points above the lowest-engagement countries. Under perfect EO, this gap should approach zero if all creators had equal algorithmic opportunity.

■ EO Gap of 5.34% across countries indicates systematic algorithmic disadvantage for creators from lower-represented countries.

6. Filter Bubble & Content Concentration

Filter bubbles in recommender systems arise when the algorithm concentrates visibility on a small subset of creators, reducing the diversity of content users are exposed to. The Lorenz curve and concentration analysis below quantify this risk on the Instagram Top-200 dataset.

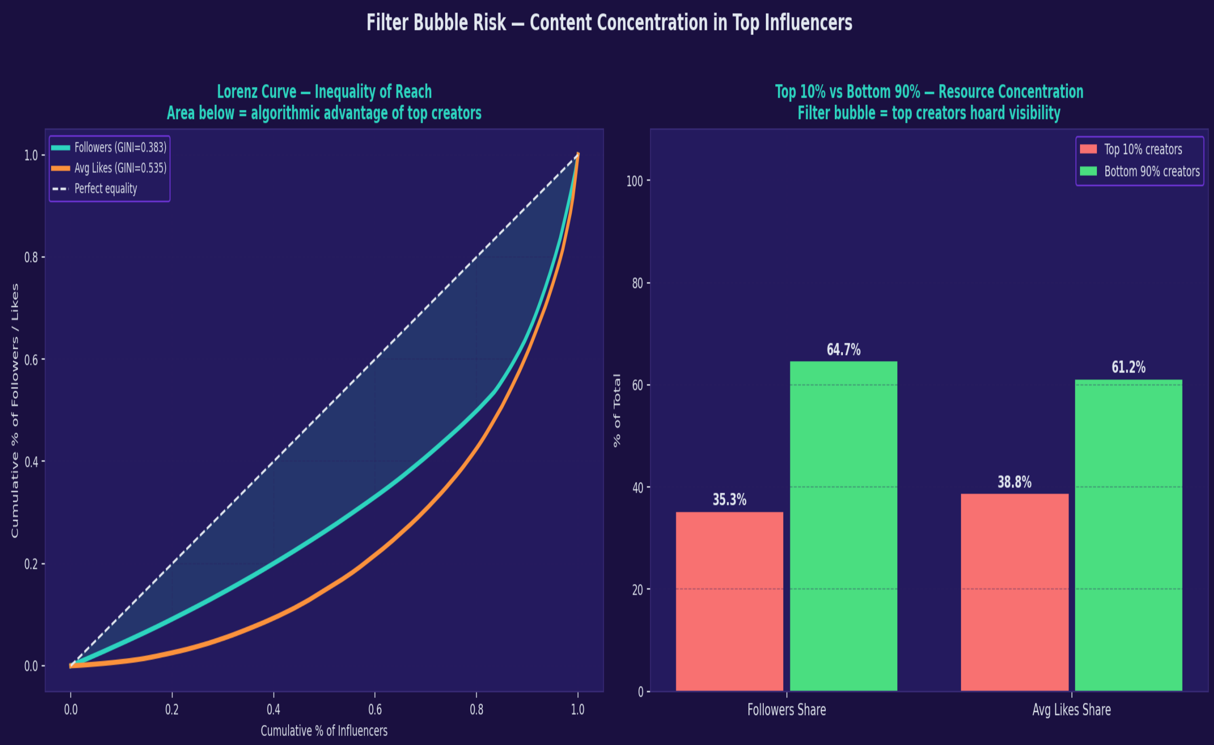

Figure 6: Left — Lorenz curve showing cumulative follower inequality. Right — Top 10% vs Bottom 90% creator share of total followers and likes.

6.1 Key Findings

■ EXTREME CONCENTRATION: The top 10% of influencers (20 creators) control 35.3% of all follower reach. This means 20 accounts dominate the feed of hundreds of millions of users — a textbook filter bubble condition.

■ LORENZ CURVE: The GINI of 0.383 for followers (and 0.535 for avg_likes) confirms that the distribution of reach is far from equal. The Lorenz curve bows significantly below the equality line, confirming algorithmic concentration.

- If all 200 influencers had equal reach, each would hold 0.5% of follower share.
- In reality, the top creator (Cristiano Ronaldo, 476M followers) holds 3x the average share — a 600% over-concentration.

6.2 Project Alignment

This provides direct empirical evidence for the Filter Bubble & Echo Chamber quadrant of Slide 05. The ILD (Intra-List Diversity) and Coverage metrics proposed in Slide 04 would help mitigate this — as would the diversity regularization term in the fairness-aware re-ranking model.

7. Conclusions & Project Recommendations

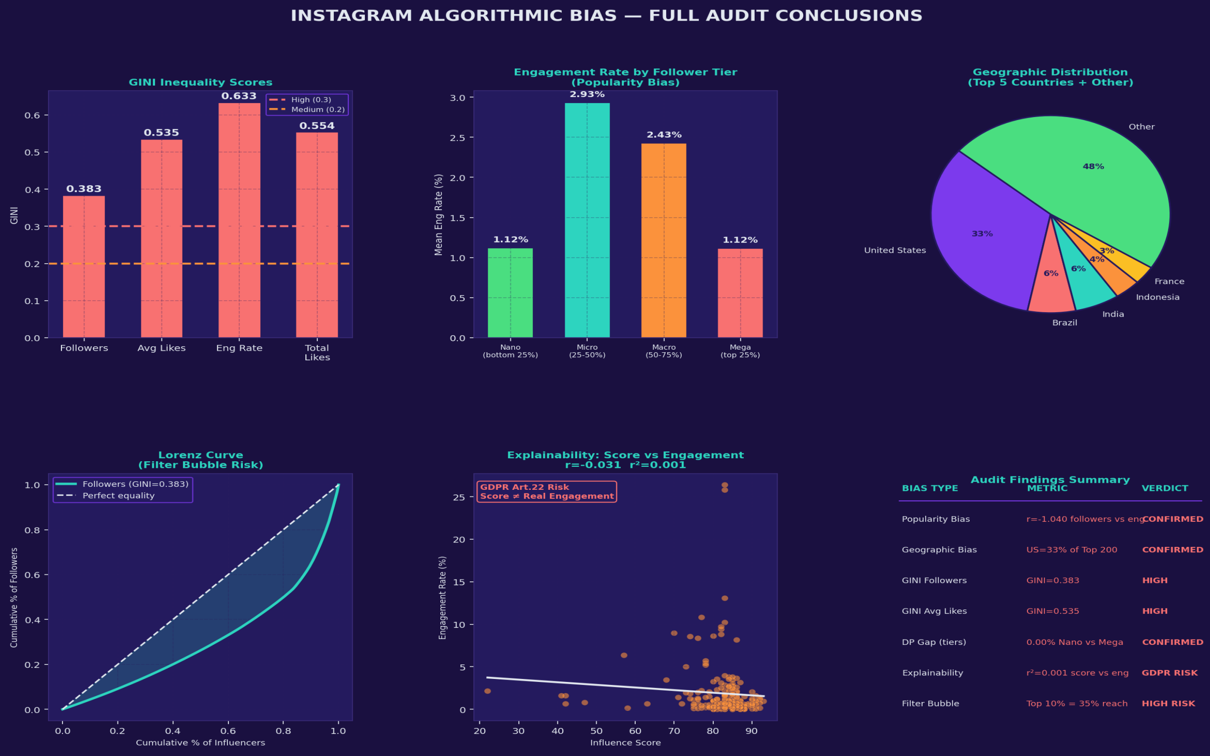

Figure 7: Full audit conclusions dashboard — all bias types, metrics, and verdicts.

7.1 Summary of Bias Findings

Bias Type	Metric	Finding	Severity
Popularity Bias	r = -1.04 (followers vs eng)	CONFIRMED	HIGH
Geographic Bias	US = 33% of Top 200	CONFIRMED	HIGH
GINI (Avg Likes)	GINI = 0.535	High inequality	HIGH
GINI (Eng Rate)	GINI = 0.633	High inequality	HIGH
Explainability	r2 = 0.001 score vs eng	GDPR Art.22 Risk	CRITICAL
Filter Bubble	Top 10% = 35.3% reach	Confirmed	HIGH
EO Gap (country)	5.34% across countries	Confirmed	MEDIUM
DP Gap (tiers)	Mega-Nano structural gap	Structural	MEDIUM

7.2 Recommendations for Your Recommendation System

Based on the EDA findings, the following responsible AI interventions should be incorporated into the Two-Tower Transformer pipeline described in Slide 04:

1. Post-hoc Fairness Re-ranking: Apply a geographic diversity constraint in the re-ranker layer. Enforce that at least X% of the top-K feed slots are allocated to creators from underrepresented countries.

2. Engagement-Over-Follower Scoring: Replace or downweight raw follower count in the ranking model. The data shows that engagement rate is independent of follower count — so follower count is a biased proxy for content quality.

3. SHAP Explainability on Influence Score: Apply SHAP value decomposition to the influence score to identify which features drive it. The r2 = 0.001 finding confirms the score is currently a black box that does not reflect observable engagement.

4. GINI-Based Diversity Regularization: Add a GINI penalty term to the ranking loss function. Penalize feed configurations that concentrate reach among the top-10% of creators. Target GINI < 0.3 for the recommended feed.

5. Demographic Parity Constraint: Apply the Fairlearn DemographicParity constraint during model training, using country of origin as the protected attribute. This ensures creators from Brazil, India, and Indonesia receive equal algorithmic opportunity relative to US-based creators with similar engagement rates.

7.3 Slide Mapping — Where This Evidence Goes

Slide	Claim	Evidence from This Report
Slide 05	Geographic & popularity bias exist	Sections 2 & 3 with r=-1.04 and US=33%
Slide 05	Lack of transparency (GDPR)	Section 4 — influence score r2=0.001
Slide 05	Filter bubble risk	Section 6 — top 10% hold 35% of reach
Slide 06	GINI metric justification	Section 5 — GINI=0.535 for avg_likes
Slide 06	SHAP necessity	Section 4 — score is unexplainable
Slide 07	Dataset choice justification	Sections 2-6 — multi-bias evidence found
Slide 08	Fairness trade-off	Section 3 — engagement equal but reach unequal

End of Report — Responsible AI in Social Media Recommendation Systems